

Practical No: - 4

ON

Cryptography and Network Security - SSCS 4020

TITLE: Implementation of Vigenère Cipher using Python

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)

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Problem Statement – 1

Aim: To implement the Vigenère Cipher algorithm for encrypting and decrypting a plaintext message using a keyword.

Theory

The **Vigenère Cipher** is a **polyalphabetic substitution cipher** that encrypts alphabetic text using a repeating keyword. Unlike the Caesar Cipher, which uses a single shift for all letters, the Vigenère Cipher uses different shifts based on the letters of the keyword.

INPUT :

Vigenere Cipher - Encryption and Decryption

```
def generate_key(text, key):  
    key = key.upper()  
    text = text.upper()  
  
    key_list = list(key)  
  
    while len(key_list) < len(text):  
        key_list.extend(list(key))  
  
    return "".join(key_list[:len(text)])  
  
def encrypt(plaintext, key):  
    plaintext = plaintext.upper().replace(" ", "")  
    key = generate_key(plaintext, key)  
  
    cipher = ""  
  
    for p, k in zip(plaintext, key):  
        x = (ord(p) - ord('A') + ord(k) - ord('A')) % 26  
        cipher += chr(x + ord('A'))
```

```
return cipher
```

```
def decrypt(ciphertext, key):
```

```
    ciphertext = ciphertext.upper().replace(" ", "")
```

```
    key = generate_key(ciphertext, key)
```

```
    plain = ""
```

```
    for c, k in zip(ciphertext, key):
```

```
        x = (ord(c) - ord('A') - (ord(k) - ord('A')) + 26) % 26
```

```
        plain += chr(x + ord('A'))
```

```
    return plain
```

```
# Main Program
```

```
plaintext = input("Enter Plaintext : ")
```

```
keyword = input("Enter Keyword : ")
```

```
ciphertext = encrypt(plaintext, keyword)
```

```
print("\nEncrypted Text :", ciphertext)
```

```
decrypted = decrypt(ciphertext, keyword)
```

```
print("Decrypted Text :", decrypted)
```

OUTPUT:

```
Enter Plaintext : INSTITUTE OF COMPUTER APPLICATION
Enter Keyword : PPSU

Encrypted Text : XCKNXIMNTDXWDBHOITJUEEDCRPLCDC
Decrypted Text : INSTITUTE OF COMPUTER APPLICATION
```

Problem Statement – 2

Aim: To simulate secure communication using the Vigenère Cipher with a shared secret key.

Theory

In secure communication, both the sender and receiver possess the same **secret keyword**.

- Sender encrypts the confidential message.
- Receiver decrypts the received ciphertext using the same keyword.

Only someone with the correct keyword can recover the original message.

INPUT :

Secure Communication using Vigenere Cipher

```
def generate_key(text, key):
```

```
    key = key.upper()
```

```
    text = text.upper()
```

```
    key_list = list(key)
```

```
    while len(key_list) < len(text):
```

```
        key_list.extend(list(key))
```

```
    return "".join(key_list[:len(text)])
```

```
def encrypt(message, key):
```

```
    message = message.upper().replace(" ", "")
```

```
    key = generate_key(message, key)
```

```
    cipher = ""
```

```
    for m, k in zip(message, key):
```

```
        x = (ord(m) - ord('A') + ord(k) - ord('A')) % 26
```

```
cipher += chr(x + ord('A'))

return cipher

def decrypt(cipher, key):
    cipher = cipher.upper().replace(" ", "")
    key = generate_key(cipher, key)
    message = ""
    for c, k in zip(cipher, key):
        x = (ord(c) - ord('A') - (ord(k) - ord('A')) + 26) % 26
        message += chr(x + ord('A'))

    return message

print("----- Sender -----")
message = input("Enter Message : ")
keyword = input("Enter Shared Secret Key : ")
cipher = encrypt(message, keyword)
print("\nCiphertext Sent :", cipher)
print("\n----- Receiver -----")
received = decrypt(cipher, keyword)
print("Received Ciphertext :", cipher)
print("Decrypted Message :", received)
```

OUTPUT:

```
----- Sender -----
Enter Message : Cryptography & Network Security
Enter Shared Secret Key : ICA

Ciphertext Sent : KTYXVOOTAXJYHPEBYOZMSMEUZKTG

----- Receiver -----
Received Ciphertext : KTYXVOOTAXJYHPEBYOZMSMEUZKTG
Decrypted Message : CRYPTOGRAPHYZNETWORKSECURITY
```

